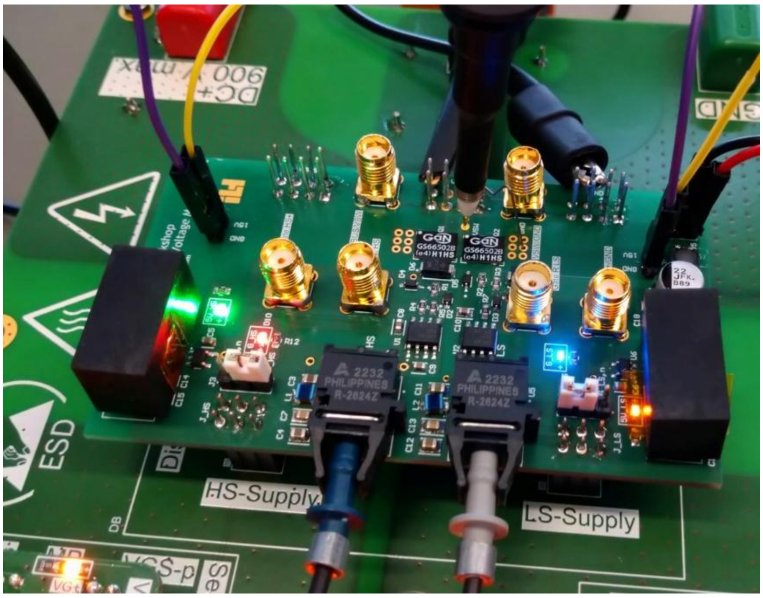
Dynamic voltage measurement of a GaN based fast switching DC / DC converter

Responsible for the experiment

Michael Bosch
Prof. Dr.-Ing. Ingmar Kallfass

Illustration of the experiment /device



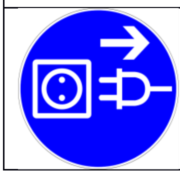
Safety instructions

The metrological experiment may only be carried out
• Individuals trained in electrical engineering
or
• under supervision by the person in charge of the experiment
> No settings may be altered or switching operations carried out on the devices without prior agreement and approval by the person responsible for the experiment (supervisor).
> There is a risk of injury from electric shock when touching exposed potential surfaces

Behaviour in case of accidents; First aid

- Note intrinsic safety
- Press the EMERGENCY STOP button and switch off the power
- Rescue injured persons, administer first aid.
- The local safety instructions, rescue plans and first aid instructions are also to be noted
- **First aider:** Mr. Bozic

Maintenance



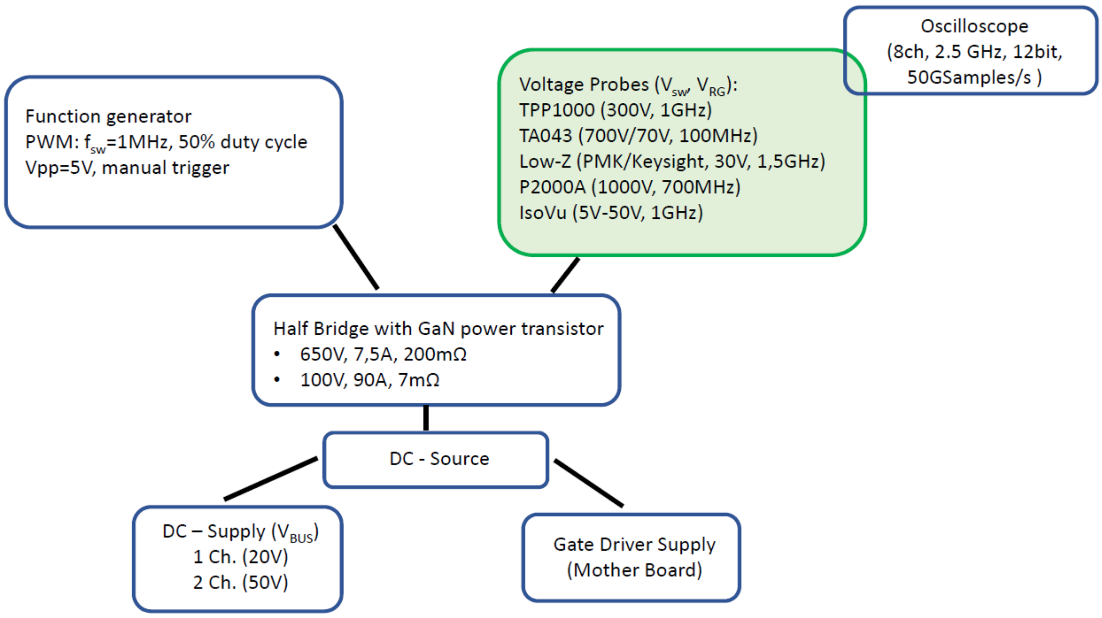
- Report defects and faults to the device owner, instructor or supervisor.
- Maintenance and repair work must be performed only by authorized personnel.

Illustration and description of the experiment and used devices

Objective of the experiment:

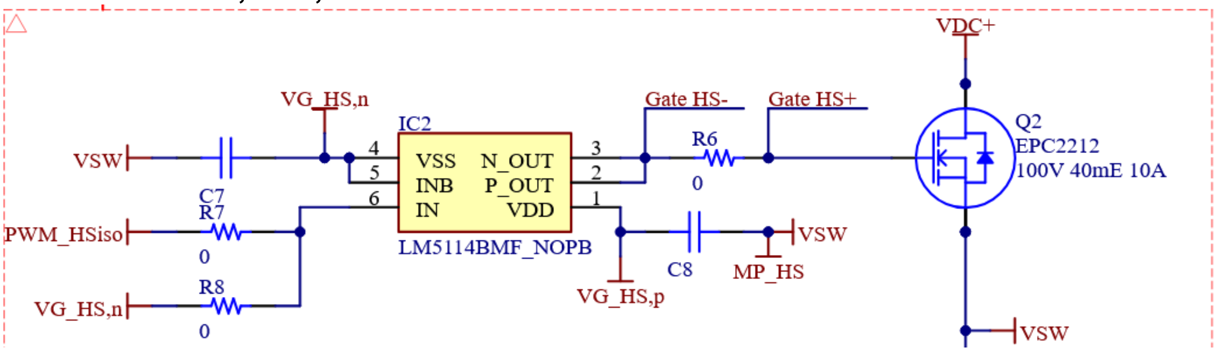
- Measurement of the switching node voltage of a GaN-based DC/DC converter
- Determination of typical switching characteristics (rise/ fall time, overshoot)
- Distinction between real switching characteristics and measuring artifacts
- Analyse the bandwidth requirement for the probes and the oscilloscope
- Measurement of the voltage across gate resistor and determination of the gate charge of high/low-side transistor

Sketch of the experimental setup

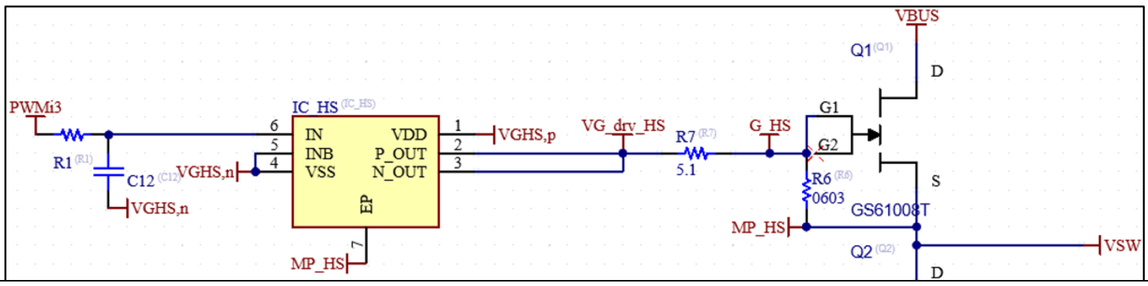


Gate Driver circuits:

EPC 2212: 100V, 18A, 13.5mΩ



GS61008T: 100V, 90A, 7mΩ



The experimental procedure / work instructions

a. Preparation for measurement

The following probes are used:

Probe type	Probe model	Specifications	Image
Passive Probe High-Z/ High- Voltage	TPP1000	300 V_{rms} 1GHz 10:1 10MΩ into 1MΩ // <4pF	
Passive Probe Low-Z / Low-Voltage	PMK PML791-RO	30 V_{rms} 1,5GHz 100:1 5KΩ into (50Ω)	
Passive Probe High Voltage	P2000A GaN Probe	1000V_{rms} 700MHz 100:1 50MΩ, 7pF	
Differential Probe	IsoVu TIV-M1/P1	5V – 50V 1GHz	
Differential Probe High-Z / High-Voltage	Pico TA043 or PMK BumbleBee	700 / 70 V_{rms} 100Mhz 100:1 /10:1 4MΩ into 1 MΩ // 7pF	

Experiment 1/2: Measurement of the voltage at the switching node with different probes

a. Experiment preparation

1. Check wire connections of the Motherboard
2. Set DC source - channel 1 for "Driver supply" and switch it on
3. Check/execute function generator settings and switch it on

Set the DC source - channel 2 for "DC-link supply" **but do not switch on, yet**

c. Performing the measurement

Procedure for a measurement:

1. Check that the function generator + DC source for driver supply is switched on
2. Check that the DC source for V_{BUS} is switched off
3. First, use the differential probe
4. Position the probe on DUT at the switching node V_{SW} and GND
5. Set the channel input impedance according to the probe specification (1MΩ/ 50Ω)
6. Enable oscilloscope channel of the probe
7. Set trigger source to the channel of the probe
8. Set trigger signal to rising edge, set trigger voltage accordingly (e.g., 15 V)
9. Start oscilloscope: “Run” or “Single”
10. Switch on DC source V_{BUS} (25V)
11. Trigger the frequency generator
12. To stop oscilloscope: Press “Stop”
13. Switch off DC source V_{BUS}
14. Only remove the probe
15. Extract characteristic values
    1. Definition see below
    2. Enter in the table below
16. Repeat steps 4-13 for the other two probes

Attention: Pay attention to the order in which the DC sources are switched on to avoid short circuits with the probe!

Note: There are two test points (plated vias) available to measure the switching node voltage of the GaN half bridge

d. Measurement evaluation

Exemplary measurement evaluation:

1. t_R / t_F : Rise / fall time, 10%-90% [definition IEEE]
2. V_{PK+} / V_{PK-}: Peak Overshoot / undershoot (min/max)
3. Oscillation: Qualitative description of the visible oscillation after the Switching process (to compare probe behaviour)

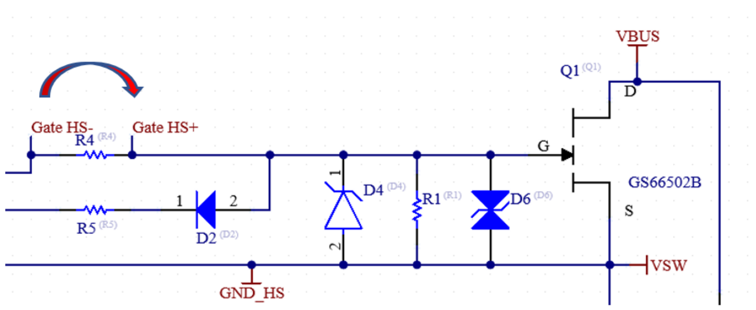
Probe	**t_R [ns]**	**V_{PK+} [V]**	**t_F [ns]**	**V_{PK-} [V]**	**Ringing Quality**
Differential probe Pico High-Z					
Passive probe with spring clip High-Z					
Passive probe with low-Z					
Passive Probe High Voltage					

Discussion of the results in the report:

1. Different measurement results
   a. Critical questioning of measurement results
   b. Observation differential probe: long-lasting and very strong ringing was measured. These are not actually in the circuit but only occur through the measuring system. (Lead probe oscillates)
   c. Compare the determined values of switching times / voltages
2. Load probe
   a. Maximum voltage at the oscilloscope input
      i. Dependent on channel input resistance/ used probe
      ii. Can be reduced by AC coupling in some instances
   b. Power loss in the probe
      i. High-impedance probe generates less power dissipation internally (U^2/R)
      ii. Low-impedance probe (low-Z) has higher power loss and is therefore limited in the voltage to be measured
      iii. Differential + passive → almost no losses → high voltage measurable
      Low-Z → significantly higher losses → only low voltage measurable, but higher bandwidth

Experiment 2/2: Measurement of the voltage across gate resistor and determination of the gate charge of the integrated high and low-side transistors

The goal of this experiment is to determine the gate charge of the low / high-side transistor in a half-bridge configuration.



Any measurement of the high side transistor basically requires the usage of an active differential probe, since the signals aren't referenced to ground potential anymore. Additionally, a determination of the gate charge is not possible with conventional differential probes, since the rapidly changing common mode voltage can disrupt the measurement. Therefore, the measurement requires a very high common mode rejection ratio (CMRR) and bandwidth.
More information at https://de.tek.com/isolated-measurement-systems
The SMD gate series resistors have a value of 1Ω or 3,3 Ω and a deviation of 1%.
The measured voltage can be used to determine the gate charge Q_G during the turn-on and turn-off transition.

$$U = R * I \rightarrow I = \frac{U}{1\Omega} \rightarrow \int I \frac{d}{dt} = Q_G$$

2.1 Experiment preparation

1. Wiring as in the first part of the experiment but with second DUT board.
2. Settings remain the same
3. Do not switch on the DC voltages of the driver and power unit yet
4. Locate the HS-Gate MMCX socket on the board and connect the IsoVu
5. Attach the TPP1000 Passive Probe to the switch node
6. Set VGS+ to 5V and VGS- to 0V

2.2 Measuring execution

Perform the measurement at a voltage of 15V, 20V and 30V.
Procedure for a measurement:

1. Check that function generator + DC source for driver supply is switched on
2. Check that DC source V_{BUS} is switched off
3. Set oscilloscope channels and the trigger conditions (edge type, trigger level)
4. Switch on DC source V_{BUS}
5. Measure and extract the characteristic values in the report
    1. Integrate (integrate voltage curve at the gate in combination with the resistor value)
6. Repeat procedure for the other gate load

2.3 Measurement analysis

EPC2202

Voltage	$Q_{\mathrm{g,HS}}$	$Q_{\mathrm{g,LS}}$
15 V		
20 V		
30 V		

GS61008T

Voltage	$Q_{\mathrm{g,HS}}$	$Q_{\mathrm{g,LS}}$
15 V		
20 V		
30 V		

In the measurement, the capacity of the single-ended probe is also measured as an error, which is 1pF.